

ADMET-Bench: Task-Dependent Performance of GNNs and Pretrained Models in ADMET Prediction

Martin Stamenov*, Mila Gjurovska*, Viktorija Vodilovska[†] and Ilinka Ivanoska*

* Faculty of Computer Science and Engineering, Ss Cyril and Methodius University, Skopje, N. Macedonia

[†] Graduate School of Life Sciences, Utrecht University, Utrecht, The Netherlands

martin.stamenov@students.finki.ukim.mk, mila.gjurovska@students.finki.ukim.mk, v.vodilovska@students.uu.nl, ilinka.ivanoska@finki.ukim.mk

Abstract—The reliability of structure-based prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties varies substantially across endpoints, yet many studies evaluate molecular models without accounting for endpoint difficulty. This work investigates the task-dependent performance of graph neural networks (GNNs), frozen pretrained molecular representations, and fingerprint-based baselines under scaffold-split evaluation, which better reflects generalization to novel chemical series. Using a unified experimental pipeline, we compare five model families across six Therapeutics Data Commons benchmarks covering four pharmacokinetic regression tasks and two toxicity classification tasks. We additionally evaluate seven hyperparameter optimization (HPO) algorithms under a fixed computational budget. Results reveal substantial differences in endpoint learnability. For structure-driven tasks such as hERG toxicity and Caco-2 permeability, GNNs achieve the strongest performance (hERG AUC = 0.825 for the best HPO trial, with a more conservative multi-seed median of 0.796). For the moderately structure-driven Tox21 NR-AR benchmark, GNNs also perform best (AUC = 0.742 for the best HPO trial, with a multi-seed median of 0.710). In contrast, for pharmacokinetic endpoints including hepatocyte clearance and half-life, all evaluated models produce near-zero or negative predictive accuracy, suggesting that molecular structure alone provides limited predictive signal. Across datasets, no HPO method consistently outperforms random search at a 50-trial budget. These findings indicate that ADMET prediction should be treated as an endpoint-specific problem rather than a single modeling task and highlight the importance of scaffold-based evaluation for realistic assessment of molecular machine learning models.

Index Terms—graph neural networks, ADMET prediction, molecular property prediction, scaffold-split, pretrained models, drug discovery, benchmarking

I. INTRODUCTION

PREDICTING molecular ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) properties is a central task in computational drug discovery, enabling early identification of promising candidates and flagging safety liabilities before costly synthesis and *in vivo* testing [1]. Graph Neural Networks (GNNs) have become prominent in this domain by treating molecules as graphs in which atoms serve as nodes and bonds as edges, enabling task-specific learning

through message passing that encodes both local chemical motifs and global topological patterns inaccessible to fixed-length descriptors [2]–[4].

Beyond GNNs, a diverse range of AI and machine learning approaches has been applied to ADMET prediction. Pretrained molecular encoders such as ChemBERTa [16] and MolCLR [13] leverage self-supervised learning on large molecular corpora to produce reusable molecular representations for downstream tasks. Classical fingerprint-based methods, including Morgan (ECFP) fingerprints [18] paired with traditional ML or MLP predictors, remain competitive baselines on many benchmarks [5], [7]. Although these approaches differ substantially in architecture and encoding strategy, all evaluated model families in this study rely exclusively on molecular structure as input, representing the same underlying 2D molecular graph or SMILES string [19] through different feature representations. GNNs operate directly on molecular graphs, fingerprint methods encode local substructures into fixed-length vectors, and transformer-based models learn contextual embeddings from SMILES sequences. Across these paradigms, strong predictive performance has been reported in the literature, often achieving state-of-the-art results on individual tasks [20].

Despite this progress, model selection in practice remains unclear. Different models perform inconsistently across tasks and molecular structures, and most published studies optimize performance for individual architectures or compare methods under different splits and experimental conditions, making direct comparison unreliable [3], [21]. Crucially, many works rely on random data splits, which allow structurally similar molecules to appear in both training and test sets. This leads to overly optimistic performance estimates that do not reflect the real-world scenario of predicting properties for new chemical scaffolds, where distribution shift is inevitable. As a result, practitioners still lack consistent guidance on model selection, endpoint-specific reliability, and the conditions under which molecular structure alone provides sufficient predictive information [20], [21].

Evaluation protocols fundamentally influence conclusions about predictive performance in molecular machine learning, particularly in settings where generalization to unseen chemical scaffolds is required. However, the extent to which evaluation protocol alters both model and optimizer rankings remains insufficiently understood. Scaffold-based split-

Ilinka Ivanoska is the corresponding author.

Martin Stamenov and Mila Gjurovska contributed equally.

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TABLE I

SUMMARY OF REPRESENTATIVE MOLECULAR PROPERTY PREDICTION METHODS. PERFORMANCE IS REPORTED FOR SELECTED TDC ADMET TASKS UNDER SCAFFOLD-SPLIT WHERE AVAILABLE. GNN = GRAPH NEURAL NETWORK, FP = FINGERPRINT, TF = TRANSFORMER, PT = PRETRAINED.

Method	Type	Key Approach	Ref.
<i>Fingerprint-Based Methods</i>			
MapLight	FP+ML	Morgan FP with RF and SVM on node embeddings	[5]
FCA	Ensemble	Weighted combination of RF, XGBoost, and SVM	[6]
RDKit2D + MLP	FP+ML	200 physicochemical descriptors with MLP	[7]
<i>Graph Neural Networks</i>			
AttentiveFP	GNN	GCN/GAT layers with GRU-based attention readout	[4]
GCN	GNN	Multi-layer graph convolution with mean pooling	[8]
GIN	GNN	Graph isomorphism layers for WL-test expressivity	[9]
D-MPNN	GNN	Directed message passing on bonds	[3]
MPNN++	GNN	Enhanced message passing (~10M params)	[10]
<i>Pretrained GNN Models</i>			
Attr. Masking	GNN+SSL	Node/edge attribute masking for pretraining	[11]
GraphCL	GNN+SSL	Graph-level contrastive learning with augmentations	[12]
MolCLR	GNN+SSL	Molecular contrastive learning of representations	[13]
MolE	GNN+FM	Multi-task pretrained GNN encoder	[14]
DeepMol	GNN+FM	Deep molecular representation learning	[15]
MiniMol	GNN+FM	Compact pretrained molecular model using MPNN++	[10]
<i>Pretrained Transformer Models</i>			
ChemBERTa-2	TF	RoBERTa pretrained on 77M SMILES (ZINC)	[16]
MolBERT	TF	BERT with SMILES and physicochemical objectives	[17]

ting [22], which partitions compounds by core ring structures, serves as a realistic evaluation protocol that reflects prospective drug discovery campaigns where models must generalize to novel chemical series. In this work, we investigate how the predictive performance of molecular AI models depends on the underlying task and evaluation protocol, with a focus on identifying when molecular structure alone provides sufficient signal. This issue is particularly important for pharmacokinetic endpoints such as clearance and half-life, where inaccurate predictions can affect compound prioritization before experimental validation. Models that appear accurate under weak evaluation protocols may therefore provide misleading support for early-stage screening decisions.

For this reason, benchmarking should identify not only which model performs best, but also which endpoints are predictable from molecular structure under realistic scaffold-shift conditions. This motivates endpoint-specific model selection, where the modeling approach is chosen according to the biological and experimental characteristics of the endpoint. From a biomedical informatics perspective, ADMET predictors act as decision-support components in early-stage drug-discovery pipelines that inform AI-supported compound prioritization. If such models are evaluated under overly optimistic protocols, they may produce overconfident predictions for novel chemical scaffolds and misguide prioritization. Understanding endpoint-specific reliability under scaffold shift is therefore essential for trustworthy AI in translational drug-discovery workflows, shifting the focus from maximizing aggregate benchmark scores toward assessing when an ADMET model is reliable enough to support compound-prioritization decisions and when its predictions should be treated as low-confidence evidence.

The main contributions of this study are as follows:

- We present a systematic benchmark of graph-based, fingerprint-based, and pretrained molecular representation

models across six ADMET endpoints from the TDC benchmark.

- We evaluate model performance under scaffold-split conditions, enabling a more realistic assessment of generalization to structurally distinct compounds.
- We compare model behavior across endpoints with different biological characteristics, showing how predictive performance varies between permeability, toxicity, cardiotoxicity, clearance, and half-life tasks.
- We assess the effect of hyperparameter optimization strategies on ADMET prediction and examine whether optimization consistently improves performance across model families and endpoints.
- We identify endpoint-specific cases where molecular structure provides sufficient predictive signal, as well as cases where structure-only representations remain limited, motivating the use of biological or assay-level information in future ADMET models.

The remainder of this paper is organized as follows. Section II reviews related work on ADMET prediction, molecular representation learning, scaffold-split evaluation, and hyperparameter optimization for molecular machine learning. Section III describes the materials and methods, including the selected TDC datasets, molecular representations, model families, scaffold-split protocol, hyperparameter optimization strategies, and evaluation metrics. Section IV presents the experimental results across regression and classification endpoints and analyzes model-family and optimizer performance. Section V discusses the implications of endpoint-dependent model behavior, scaffold-split generalization, and the limitations of structure-only molecular representations. Finally, Section VI concludes the paper and outlines directions for future work.

II. RELATED WORK

A. ADMET Prediction

The TDC ADMET benchmark [20] has become the standard evaluation platform for molecular property prediction. Table I summarizes representative approaches from the TDC leaderboard and the broader literature.

The TDC leaderboards [23] reveal that several top-performing approaches rely on comparatively simple architectures. MapLight [5] uses random forests and SVMs on node embeddings, while RDKit2D+MLP [7] passes 200 physicochemical descriptors through a standard multi-layer perceptron. On certain datasets, even Random Forest or XGBoost achieves top results, motivating ensemble approaches such as FCA [6], which combines multiple classifiers via weighted voting.

Among GNN-based methods, AttentiveFP [4] employs GCN/GAT layers with GRU-based attention readout, achieving top-5 results on many TDC datasets. Standard architectures appearing prominently in molecular benchmarks include GCN [8], GIN [9], and D-MPNN [3], which operates on bond-level rather than atom-level messages.

Self-supervised pretraining has been adapted from NLP to molecular graphs via two main paradigms: attribute masking [11], which reconstructs removed node or edge features, and contrastive learning [12], which trains encoders to produce similar embeddings for augmented views of the same molecule. MolCLR [13] extends contrastive learning with three graph augmentation strategies (atom masking, bond deletion, subgraph removal), demonstrating improved downstream ADMET prediction compared to training from scratch.

These techniques underpin most pretrained molecular representation models, which follow a common paradigm: self-supervised pretraining on large unlabeled corpora, optional multi-task supervised training, and task-specific fine-tuning. Representative examples include DeepMol [15] and MoLE [14]. ChemBERTa [16] takes a distinct approach, applying a RoBERTa transformer to SMILES strings pretrained on 77M molecules from ZINC-15 [24], while MolBERT [17] combines SMILES masked language modeling with auxiliary physicochemical objectives. A practical concern with pretrained molecular representation models is their large parameter count; MiniMol [10] addresses this by using MPNN++ (~10M parameters) to achieve competitive TDC performance at substantially lower computational cost.

a) Scaffold Split Challenges: Transformer-based models frequently show severe generalization drops under scaffold splitting, with strong validation scores that collapse toward chance-level performance on held-out scaffolds [3], [16]. Wu et al. [21] introduced MoleculeNet with scaffold-splitting to more faithfully reflect prospective drug discovery generalization. This observation is revisited in Section IV, where the experimental results show that on Tox21 NR-AR, ChemBERTa achieves validation AUC = 0.896, but drops to 0.73 on the scaffold-split test set, while MolCLR falls below the random baseline on the test set (AUC = 0.45).

B. Hyperparameter Optimization for Molecular ML

HPO is a key performance driver in molecular ML, where small, noisy datasets amplify sensitivity to hyperparameter configuration. Bergstra and Bengio [25] established Random Search as a strong baseline, showing it outperforms grid search when only a subset of hyperparameters is influential. Bayesian methods such as TPE [26], implemented in Optuna [27], improve sample efficiency by modeling the hyperparameter-performance relationship, making them attractive under limited trial budgets.

Population-based metaheuristics, including PSO [28], GA [29], and ABC [30], provide broader exploration, but may require larger budgets to converge. SA [31] offers a middle ground by accepting suboptimal moves early to escape local optima. Despite their widespread use in combinatorial optimization, systematic comparisons of these HPO strategies for molecular property prediction remain scarce, particularly under scaffold-split protocols, motivating our evaluation of multiple optimization approaches within a unified ADMET benchmark.

C. Challenges in Molecular ML: Distribution Shifts and Generalization

Beyond the scaffold-split sensitivity discussed above, molecular property prediction faces additional generalization challenges that compound evaluation difficulty. First, ADMET datasets are typically small (hundreds to low thousands of compounds), which amplifies variance across data splits and makes HPO-selected configurations sensitive to the specific training fold [21]. Second, the relationship between molecular structure and pharmacokinetic endpoints is often indirect: properties such as clearance and half-life depend on enzyme kinetics, transporter activity, and protein binding factors not explicitly encoded in molecular graphs or SMILES strings [32]. Third, class imbalance [33] in toxicity assays (e.g., 4.2% positive rate in Tox21 NR-AR) biases standard loss functions toward the majority class, a problem that persists in our experiments where no explicit class reweighting is applied.

These compounding factors mean that reported model rankings can shift substantially depending on dataset size, splitting protocol, and evaluation metric. Robust benchmarking therefore requires multi-seed validation, paired effect-size estimation, and confidence intervals, practices that remain inconsistent in molecular ML. Our framework enforces all three across every comparison.

III. MATERIALS AND METHODS

A. Framework Overview

To compare model families under realistic generalization conditions, we use a benchmarking framework (Fig. 1) that applies the same scaffold partitions, datasets, and metrics across the GNN and the frozen-pretrained/fingerprint pipelines. The framework coordinates dataset loading, scaffold-split partitioning, model training, metric computation, and statistical analysis. The statistical analysis includes repeated runs with different random seeds, aggregation of across-seed summary

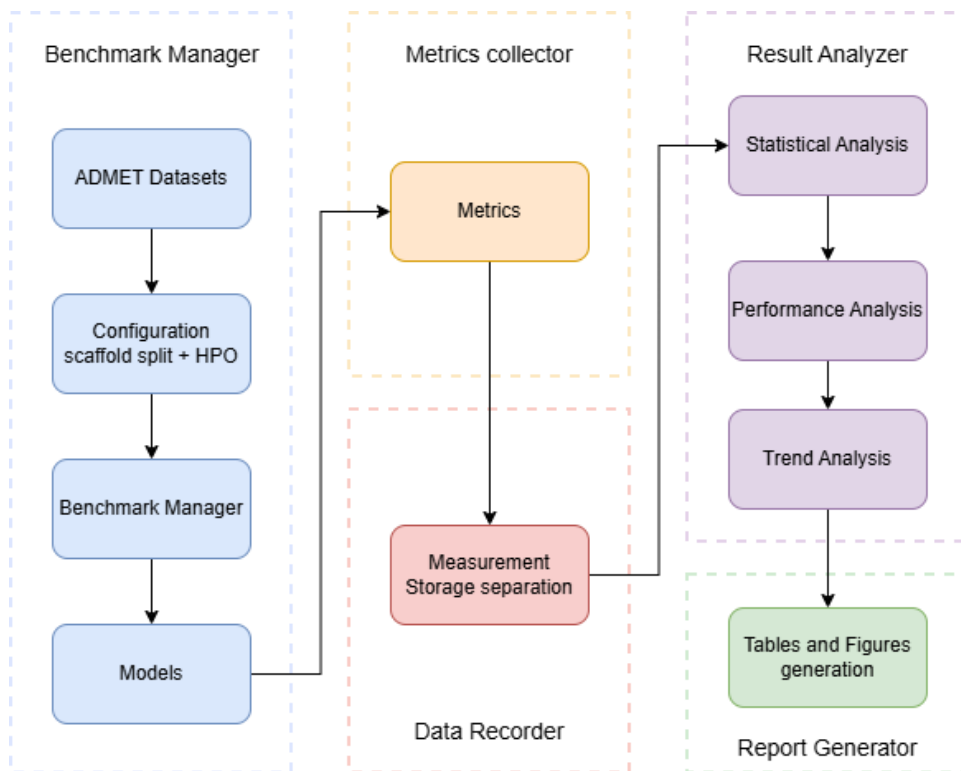


Fig. 1. Modular benchmarking architecture for evaluating molecular graph neural networks and pretrained models on ADME and toxicity prediction tasks. The framework consists of three layers: a benchmark orchestration layer coordinating datasets and experimental configurations, a model execution layer evaluating multiple neural architectures, and an evaluation pipeline.

statistics (medians and ranges), comparison of model families across endpoints, and confidence-interval-based comparison of hyperparameter optimization strategies against a Random Search baseline. Fixing the splits, datasets, and metrics across models removes split- and metric-related confounds that complicate cross-study comparison in molecular ML [21], so that observed differences reflect model characteristics rather than experimental artifacts. The design separates experiment orchestration, model execution, and evaluation into independent components, so new algorithms, architectures, or datasets can be added without modifying the evaluation logic.

B. Datasets and Evaluation Protocol

We evaluate all models on six molecular property prediction tasks drawn from the Therapeutics Data Commons (TDC) ADMET benchmark [20], covering four pharmacokinetic (ADME) regression tasks and two toxicity classification tasks. This evaluation protocol reflects practical drug discovery scenarios, where models are required to generalize to novel chemical series. Fig. 2 provides an overview of data provenance and class composition across all six datasets.

1) ADME Regression Tasks:

a) Caco-2 Permeability (Caco2_Wang): The Caco-2 cell monolayer assay is the gold-standard *in vitro* surrogate for human intestinal absorption [34]. High apparent permeability ($P_{app} > 10^{-6}$ cm/s) is associated with good oral bioavailability. The dataset, curated by Wang et al. [35], contains **910 compounds** with target values expressed in log(cm/s) units.

Caco-2 showed the strongest regression performance in our benchmark, likely because permeability is closely related to molecular size, lipophilicity, and hydrogen-bonding capacity, which are represented directly in molecular graphs.

b) Half-Life Prediction (Half_Life_Obach): Plasma half-life ($t_{1/2}$, hours) determines dosing frequency and is essential for designing drugs with adequate therapeutic windows. However, half-life is a composite pharmacokinetic parameter influenced not only by molecular structure, but also by volume of distribution, protein binding fraction, and individual patient variability [32]. The dataset is drawn from the Obach FDA-approved drug collection and contains **667 compounds** spanning a wide dynamic range (0.5-100+ hours). We apply a log transformation before standardization to reduce target skewness.

c) Hepatocyte Clearance (Clearance_Hepatocyte_AZ): Intrinsic clearance measured in freshly isolated human hepatocytes ($\mu\text{L}/\text{min}/10^6$ cells) captures both phase-I and phase-II metabolic routes and is the most physiologically relevant *in vitro* metabolic stability assay [36]. The dataset was generated internally at AstraZeneca and is distributed through TDC [20], comprising **1213 proprietary compounds**. This dataset yielded the weakest regression performance in our benchmark, with GNN models producing negative R^2 values, suggesting that, within this benchmark, molecular structure alone was insufficient to explain the observed variance in hepatic metabolism (see Section IV).

d) Microsomal Clearance (Clearance_Microsome_AZ): Human liver microsomal assays ($\mu\text{L}/\text{min}/\text{mg}$ protein) mea-

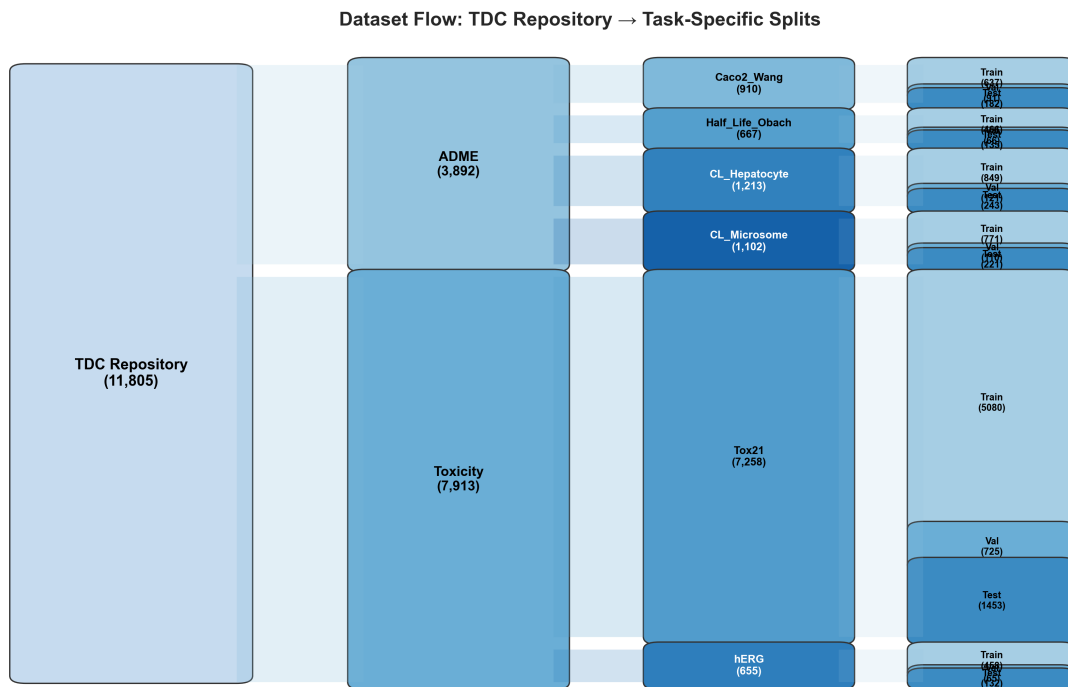


Fig. 2. Overview of data source and task labels. This flowchart details the 11,805 total instances catalogued in the TDC Repository. After removing molecules with invalid SMILES or failed graph conversion, 10,627 molecules are retained for training and evaluation.

sure primarily cytochrome P450-mediated (CYP) oxidative metabolism; they offer higher throughput than hepatocyte assays, but capture only phase-I reactions [37]. The AstraZeneca dataset contains **1 102 compounds**. Like hepatocyte clearance, microsomal clearance values are log-transformed and then standardized using train-set statistics.

2) Toxicity Classification Tasks:

a) *Tox21 Nuclear Receptor: Androgen Receptor (Tox21 NR-AR)*: The Tox21 initiative [38] assembled a large multi-assay toxicology panel to enable high-throughput environmental hazard screening. We use the **NR-AR** assay (androgen receptor agonism), which identifies compounds that can disrupt androgen signaling. The TDC catalogue lists **7 258 compounds** (6 533 after graph conversion), of which only $\approx 4.2\%$ are active (positive) samples across the full corpus (see Fig. 2), making this the most class-imbalanced task in our benchmark; the scaffold-split test set has a slightly higher positive rate of $\approx 4.9\%$.

b) *hERG Cardiotoxicity (hERG)*: Blockade of the hERG potassium ion channel is the predominant mechanism of drug-induced QT interval prolongation and potentially fatal cardiac arrhythmias, and the hERG assay is a mandatory safety screen in preclinical drug development [39]. The TDC hERG dataset contains **655 compounds** with binary blocker/non-blocker labels; approximately 69% of compounds are blockers (a roughly 69/31 class split), making this a considerably more balanced classification task than Tox21 NR-AR.

c) *Splitting and evaluation protocol*: We use the TDC scaffold-split function, which partitions compounds by Bemis-Murcko scaffolds [22] so that molecules with structurally distinct core rings appear in different folds. Scaffold-based split-

ting approximates prospective deployment conditions, where a model must generalize to structurally distinct compound families not seen during training. We further subdivide the train set 90/10 (train/validation) using a fixed random seed (42) to enable early stopping and model selection without touching the held-out test set. The final split proportions are approximately 70/8/22 (train/validation/test) across all datasets. Because scaffold splitting assigns entire scaffold groups to the same partition, the exact proportions may vary slightly across datasets.

During hyperparameter optimization (HPO), the test split remains *fully held out*; model selection is performed solely on the validation metric. Final performance is reported once on the test split after HPO is complete, preventing any implicit information leakage from repeated test-set evaluations.

d) *Target transformations*: For the three skewed regression tasks (Half-Life, Hepatocyte and Microsomal Clearance) we apply $y' = \log(\max(y, \epsilon))$, where $\epsilon = 10^{-3}$, followed by z-score normalization using training-set statistics. Caco-2 targets are already in $\log(\text{cm/s})$ units, so only z-score normalization is applied. All normalization parameters are computed exclusively on the training fold and reused for validation and test sets to prevent data leakage. Classification targets are kept as binary integers (0/1) without transformation.

C. Model Families

This comparison allows us to evaluate whether task-specific learning provides advantages over frozen pretrained molecular representations and fixed fingerprint baselines. We evaluate

three distinct model paradigms under the same scaffold splits and evaluation metrics.

a) GNN models: We employ graph neural networks trained end-to-end on each task. The GNN backbone is based on stacked GraphConv layers [40] with batch normalization, ReLU activation, and additive residual connections between successive layers, selected through a preliminary architecture comparison across eight GNN variants (Section IV-D). Each molecule is represented as a molecular graph constructed from SMILES using RDKit. Atom features include atomic number, degree, formal charge, hybridization, aromaticity, ring membership, hydrogen count, and atomic mass (8 features); no bond features are used. The GNN encoder produces node embeddings aggregated into a graph-level representation using concatenated global mean and max pooling. Before the prediction head, this pooled graph representation is concatenated with a small vector of structure-derived physicochemical descriptors computed with RDKit (15 descriptors for the ADME regression and toxicity tasks: molecular weight, cLogP, hydrogen-bond donors and acceptors, topological polar surface area, rotatable bonds, aromatic and aliphatic ring counts, molar refractivity, Bertz complexity, heteroatom count, and four Lipinski-rule flags; a 7-descriptor subset for Caco-2). These descriptors are derived from molecular structure alone and add no biological or assay information; they are provided only to the GNN and not to the pretrained or fingerprint baselines (see Section V). The combined representation is passed to an MLP prediction head with batch normalization and optional dropout. To regularize the small Caco-2 dataset, the GNN additionally applies dropout (0.5) and Gaussian target-label noise ($\sigma = 0.05$) for that endpoint only. GNN models undergo full hyperparameter optimization (Section III-D).

b) Pretrained models: We evaluate two pretrained molecular encoders representing the two major pretrained molecular model paradigms shown in Table I. We use SMILES-based transformer models such as ChemBERTa [16] and graph-based contrastive models such as MolCLR [13] as representative examples of sequence-based and graph-based pretrained approaches, rather than as an exhaustive list of all available pretrained molecular models. For pretrained encoders, we use a frozen encoder and trainable MLP head setting: embeddings are extracted once per molecule, and an MLP is trained on the train split with early stopping based on the validation split. This frozen-encoder protocol isolates the quality of pretrained representations from downstream optimization effects.

c) Fingerprint baselines: We evaluate two fingerprint-based approaches: (i) Morgan fingerprints (ECFP) [18] with an MLP predictor, and (ii) MoIE-style learned fingerprints (MoIE-FP) with an MLP predictor. The MoIE-FP baseline is implemented as Morgan fingerprints followed by a learned projection inspired by the MoIE approach [14], rather than the original pretrained MoIE encoder. These fixed molecular descriptors serve as baselines representing the classical cheminformatics approach to molecular property prediction. All baselines use the same dataset splits and evaluation metrics as the GNN experiments.

D. Model Training and Configuration

a) GNN Training: All GNN models were trained using the Adam optimizer with a batch size of 32. During HPO, models were trained for a maximum of 50 epochs per trial with early stopping (patience = 12). Regression models were optimized using mean squared error (MSE) loss, while classification models used binary cross-entropy (BCE) loss with logits. Gradient clipping was applied with a maximum norm of 1.0. Table II defines the hyperparameter search space. The three MLP-head widths are constrained to be non-increasing ($\text{Layer 1} \geq \text{Layer 2} \geq \text{Layer 3}$), so a sampled triple is sorted in descending order before use, yielding the funnel-shaped prediction head used throughout.

b) Optimization algorithms: We evaluate seven HPO strategies, Random Search [25], Particle Swarm Optimization (PSO) [28], Artificial Bee Colony (ABC) [30], Genetic Algorithm (GA) [29], Simulated Annealing (SA) [31], Hill Climbing (HC) [41], and Tree-structured Parzen Estimator (TPE) [27]. For population-based methods (PSO, GA, ABC), a population size of 16 is used. SA is initialized with temperature $T_0 = 1.0$ and exponential cooling ($\alpha = 0.99$). TPE uses Optuna’s default of 10 startup trials with median pruning. All NiaPy-based [42] methods operate over the 7-dimensional search space in Table II. TPE (Optuna) searches a separate space over hidden dimension, number of layers, learning rate, weight decay, and dropout rate $\in [0.0, 0.5]$. Each algorithm is allocated 50 trials per dataset, giving 2,100 GNN training runs from the optimizer comparison alone (seven optimizers $\times$ six datasets $\times$ 50 trials); together with the preliminary architecture-selection screen (240 runs: eight architectures $\times$ six endpoints $\times$ five configurations) and the five-seed validation (30 runs), the study comprises approximately 2,370 training runs in total.

E. Experimental Setup

a) Datasets: Table III summarizes the dataset characteristics across all six tasks.

b) Evaluation metrics: Model performance is assessed using RMSE, R^2 , and MAE for ADME regression tasks, and AUC-ROC and F1-score for toxicity classification tasks. AUC-ROC is the primary ranking metric for classification; F1 is reported in addition because it is more sensitive to the minority class under strong class imbalance, and it is the classification metric used in the paired optimizer comparison (Table IX).

c) Multi-seed validation: To quantify robustness, we retrain the best HPO-selected configuration per dataset across five seeds [42, 123, 456, 789, 1011]. We report the median and the full min–max range across seeds; the median is used because the best configuration diverges on a minority of seeds for the weakly-learnable endpoints, which makes the across-seed mean unrepresentative.

d) HPO algorithm comparison: We compare each optimizer against the Random Search baseline using paired performance differences across the six datasets rather than null-hypothesis significance tests. For every dataset we compute the improvement with a normalized direction so that positive values favor the optimizer (for RMSE, $\text{Random} - \text{Algo}$; for

TABLE II

HYPERPARAMETER SEARCH SPACE FOR GNN MODELS. THE THREE MLP-HEAD LAYER WIDTHS ARE SAMPLED INDEPENDENTLY AND THEN SORTED IN DESCENDING ORDER (LAYER 1 $\geq$ LAYER 2 $\geq$ LAYER 3) TO YIELD A FUNNEL-SHAPED PREDICTION HEAD.

Hyperparameter	Search Space
Hidden Dimensions	{64, 96, 128, 192, 256, 384, 512}
Number of Layers	{3, 4, 5, 6, 7}
Learning Rate	[1e-4, 1e-2] (log-uniform)
Weight Decay	[1e-6, 1e-2] (log-uniform)
MLP Head Layer 1	{128, 192, 256, 384, 512}
MLP Head Layer 2	{64, 96, 128, 192, 256}
MLP Head Layer 3	{32, 48, 64, 96, 128}

TABLE III

DATASET STATISTICS AND EVALUATION METRICS (TDC ADMET). “TDC” SHOWS COMPOUNDS CATALOGUED; “USED” SHOWS THOSE RETAINED AFTER FILTERING. RMSE IS THE PRIMARY PER-DATASET REGRESSION METRIC; THE SCALE-INVARIANT R^2 IS ADDITIONALLY USED FOR CROSS-MODEL COMPARISON IN TABLE IV.

Dataset	Type	TDC	Used	Split	Metric
Caco2_Wang	Regr.	910	819	Scaffold	RMSE
Half_Life	Regr.	667	601	Scaffold	RMSE
Clear_Hep.	Regr.	1,213	1,092	Scaffold	RMSE
Clear_Mic.	Regr.	1,102	992	Scaffold	RMSE
Tox21 (NR-AR)	Class.	7,258	6,533	Scaffold	AUC
hERG	Class.	655	590	Scaffold	AUC

F1, Algo – Random). Because the datasets differ by orders of magnitude in scale, we express each difference as a percentage of the Random-Search baseline before aggregating. For each optimizer we report the win/tie/loss count, the mean relative improvement, a 95% confidence interval obtained from 10,000 percentile bootstrap resamples of the six paired differences, and the paired effect size d_z (mean divided by the standard deviation of the differences). The paired unit is the dataset ($n = 6$), and all comparisons use final held-out test performance obtained after HPO model selection on the validation split. With only six paired observations, null-hypothesis significance tests are underpowered; we therefore rely on bootstrap confidence intervals and the paired effect size, which together summarize both the magnitude and the consistency of any improvement over Random Search.

e) Implementation: The implementation was carried out in Python ≥ 3.8 using PyTorch ≥ 2.0 , PyTorch Geometric ≥ 2.3 , RDKit ≥ 2022.9 , PyTDC ≥ 0.4 , NiaPy ≥ 2.0 [42], Optuna ≥ 3.0 , and Transformers ≥ 4.30 . All experiments were executed on a workstation equipped with an Intel Core i7-8700K CPU running at 3.70 GHz, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU. The total runtime required to complete all experimental runs was approximately 45 hours.

IV. RESULTS

A. Task-Level Performance on ADMET Endpoints

Table IV presents the performance of all model families across the six ADMET tasks. Fig. 3 provides a visual summary of relative model rankings.

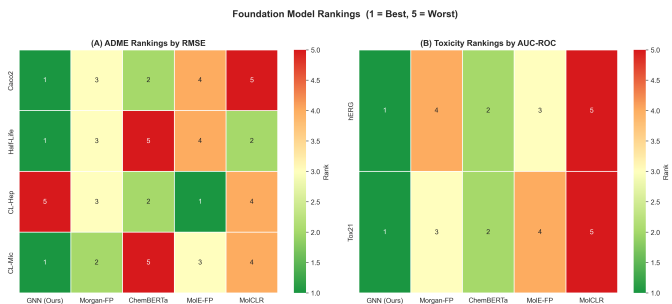


Fig. 3. Task-dependent performance of different model families across ADMET endpoints. Green indicates better performance within each task.

We observe substantial variability in predictive performance across tasks, which we organize into three groups based on the degree to which molecular structure supports prediction. Although the grouping is assigned post-hoc from the observed test performance, the tiers correspond to approximate, reusable boundaries on the best model per endpoint: structure-driven tasks reach regression $R^2 \gtrsim 0.45$ or classification AUC $\gtrsim 0.80$; moderately structure-driven tasks fall in the intermediate band ($R^2 \approx 0.1\text{--}0.3$ or AUC $\approx 0.70\text{--}0.75$); and weakly structure-driven tasks sit near or below chance ($R^2 \lesssim 0.05$). These thresholds are offered as a practical triage heuristic rather than as sharp category limits, and we expect endpoints dominated by passive, structure-determined processes (permeability, direct channel binding) to fall in the upper tiers, whereas endpoints governed by enzyme kinetics, transporter activity, or protein binding tend toward the lower tiers.

a) Group 1: Structure-driven tasks.: On hERG cardiotoxicity and Caco-2 permeability, the GNN achieves the strongest performance (AUC = 0.825 and $R^2 = 0.481$, respectively). These endpoints exhibit strong structure-activity relationships: hERG channel blockade is associated with specific pharmacophoric motifs, and intestinal permeability is governed primarily by molecular size, lipophilicity, and hydrogen-bond capacity. On both tasks, the GNN matches or exceeds all pretrained encoders and fingerprint baselines.

b) Group 2: Partially structure-driven tasks.: On Tox21 NR-AR and Microsomal clearance, the GNN achieves the best performance (AUC = 0.742 and $R^2 = 0.191$), but the margin over other model families narrows. On Tox21, all models except MolCLR achieve AUC > 0.67 , indicating that multiple representation strategies capture relevant structural signals to a similar degree. On Microsomal clearance, predictive power is marginal across all models, with GNN $R^2 = 0.191$ and the next-best Morgan-FP reaching $R^2 = 0.122$. Structure alone is partially informative for these endpoints, but additional features or data may be needed for stronger predictions.

c) Group 3: Tasks weakly learnable from structure alone.: On Half-Life and Hepatocyte clearance, all evaluated models produced weak predictive performance, with near-zero or negative R^2 values. The best Half-Life R^2 is 0.004 (GNN), effectively zero, while all other models produce negative R^2 values. On Hepatocyte clearance, the GNN yields $R^2 = -1.019$ (worse than a mean predictor), and even the best

TABLE IV

TASK-DEPENDENT PERFORMANCE OF DIFFERENT MODEL FAMILIES ACROSS ADMET ENDPOINTS. R^2 (HIGHER IS BETTER) IS REPORTED FOR REGRESSION TASKS; AUC-ROC (HIGHER IS BETTER) FOR CLASSIFICATION TASKS. BEST PER TASK IN **BOLD**. “GNN-BEST” IS THE SINGLE BEST CONFIGURATION FOUND DURING THE 50-TRIAL HPO SEARCH; BECAUSE BEST-OF-50-TRIALS SELECTION IS OPTIMISTIC, THE MULTI-SEED RE-VALIDATION MEDIAN (TABLE XI) ARE LOWER FOR THE CLASSIFICATION ENDPOINTS (HERG 0.796, Tox21 0.710) AND SHOULD BE CONSULTED FOR EXPECTED DEPLOYMENT PERFORMANCE. R^2 IS USED FOR REGRESSION COMPARISON AS IT IS SCALE-INVARIANT ACROSS DIFFERENT TARGET PREPROCESSING.

Task	Metric	GNN-Best	ChemBERTa	Morgan-FP	MolE-FP	MolCLR
<i>Group 1: Structure-driven tasks</i>						
hERG	AUC $\uparrow$	0.825	0.770	0.611	0.672	0.401
Caco-2	R^2 $\uparrow$	0.481	0.478	0.200	0.047	-0.189
<i>Group 2: Partially structure-driven tasks</i>						
Tox21 (NR-AR)	AUC $\uparrow$	0.742	0.728	0.722	0.675	0.452
Clearance Mic.	R^2 $\uparrow$	0.191	0.024	0.122	0.059	0.041
<i>Group 3: Tasks weakly learnable from structure alone</i>						
Half-Life	R^2 $\uparrow$	0.004	-0.594	-0.039	-0.329	-0.001
Clearance Hep.	R^2 $\uparrow$	-1.019	0.029	-0.015	0.032	-0.039

model (MolE-FP, $R^2 = 0.032$) explains essentially no variance. These results suggest that, for the evaluated structure-only representations under scaffold-split evaluation, molecular structure alone provided limited predictive signal for these pharmacokinetic endpoints, which depend on enzyme kinetics, transporter activity, protein binding, and individual patient variability [32].

These results suggest that molecular structure alone may be insufficient for endpoints that depend strongly on biological and experimental context. Both simple and more complex models showed similarly weak performance on the same endpoints, which is more consistent with a limitation in the information content of the structure-only representation than with the choice of learning algorithm.

B. Model Performance Across Tasks

We examine performance differences between model families in greater detail using Fig. 4, which compares GNN, pretrained, and fingerprint models across all tasks.

a) GNN advantage on toxicity tasks.: Within this experimental setup, the GNN outperforms the two frozen pretrained baselines evaluated here on both toxicity classification tasks. On hERG, the GNN achieves AUC = 0.825 compared to ChemBERTa’s 0.770 and Morgan-FP’s 0.611. On Tox21, the GNN (AUC = 0.742) matches ChemBERTa (0.728) and substantially exceeds MolCLR (0.452). Task-specific graph convolution patterns appear valuable for toxicity endpoints, where specific substructural motifs are strong predictors. We note, however, that the GNN additionally receives auxiliary physicochemical descriptors that the frozen baselines do not. To decompose this advantage we perform a controlled descriptor ablation (Section IV-C, Table V): on hERG the descriptors account for roughly 0.08 AUC of the GNN’s performance, so that without them the graph encoder alone (≈ 0.76) is comparable to, rather than clearly ahead of, frozen ChemBERTa (0.770). Part of the GNN’s headline toxicity advantage therefore reflects the auxiliary descriptor input rather than the graph representation in isolation (Section V).

b) Comparable performance on permeability.: On Caco-2, the optimized GNN achieves $R^2 = 0.48$, comparable to ChemBERTa’s $R^2 = 0.48$. Both graph-based and SMILES-based representations capture a similar degree of the underlying structure-activity relationship for permeability prediction.

c) Limited performance on clearance and half-life.: On Hepatocyte clearance, no evaluated model achieves meaningful predictive power (all $R^2 \approx 0$). The frozen pretrained representations provide less extreme, though still poor, predictions than end-to-end GNN training on this endpoint: ChemBERTa [16] achieves $R^2 = 0.03$ versus GNN $R^2 = -1.02$. This result should be interpreted as evidence that frozen representations can serve as regularized baselines in this benchmark, not as evidence that pretrained molecular models are generally superior for clearance prediction. On Half-Life and Microsomal clearance, the same pattern of weak or absent predictive signal persists across all model families.

For practitioners, these results indicate a diagnostic approach: if a task-specific GNN achieves negative R^2 or near-random AUC on a scaffold-split test set, the endpoint may not be predictable from molecular structure alone, and frozen pretrained encoders can be included as regularized baselines for comparison.

C. Descriptor Ablation

Because the GNN alone receives the auxiliary physicochemical descriptors, we run a controlled ablation to separate their contribution from that of the learned graph representation. For each endpoint we take the best HPO-selected configuration (Table VII) and retrain the GraphConv model twice under an identical training budget (50 epochs, patience 12) and identical seeds, once with the descriptors and once with them removed (the pooled graph embedding is passed directly to the prediction head). We report the median across three seeds [42, 123, 456]; because these are re-trained under a fixed shared budget rather than the full best-trial schedule, the with-descriptor column is close to but not identical to the best-trial values in Table IV.

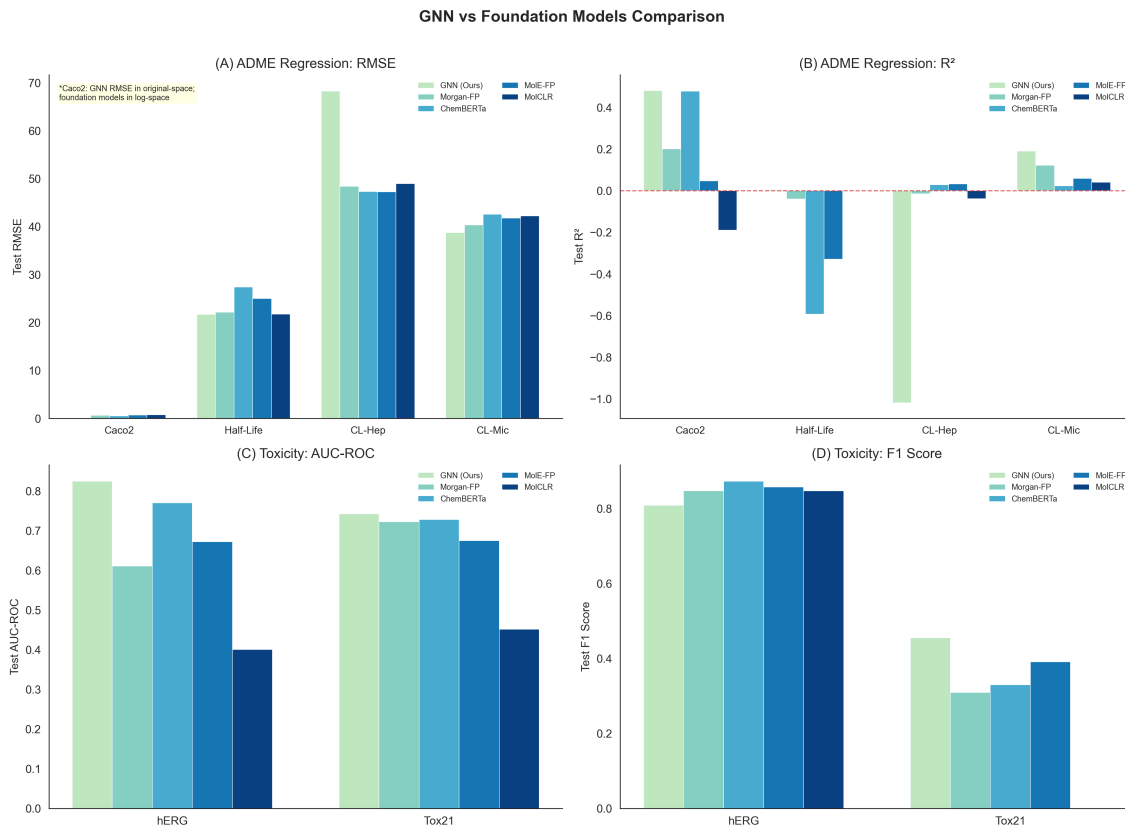


Fig. 4. Comparison of GNN (with HPO) against pretrained and fingerprint baselines across all datasets, using the metrics reported in Table IV. (A) ADME regression R^2 (higher is better; dashed line at $R^2 = 0$), (B) toxicity AUC-ROC (dashed line at the 0.5 random baseline).

TABLE V
DESCRIPTOR ABLATION FOR THE GRAPHCONV GNN: TEST PERFORMANCE WITH AND WITHOUT THE AUXILIARY PHYSICOCHEMICAL DESCRIPTORS, USING THE BEST HPO CONFIGURATION PER ENDPOINT RETRAINED UNDER A MATCHED BUDGET (MEDIAN OVER THREE SEEDS). METRIC IS R^2 FOR REGRESSION AND AUC-ROC FOR CLASSIFICATION; Δ IS THE DESCRIPTOR CONTRIBUTION (WITH – WITHOUT).

Endpoint	Metric	With desc.	Without desc.	Δ
Caco-2	R^2	0.422	−0.691	+1.11
Half-Life	R^2	0.171	0.009	+0.16
Clearance Hep.	R^2	−0.133	−0.070	−0.06
Clearance Mic.	R^2	0.122	0.064	+0.06
Tox21 (NR-AR)	AUC	0.707	0.665	+0.04
hERG	AUC	0.837	0.760	+0.08

The ablation shows that the descriptors are not a uniform bonus but are decisive precisely where the GNN appears strongest. On Caco-2 permeability, removing the descriptors collapses performance from $R^2 = 0.42$ to -0.69 : the graph representation alone does not solve this endpoint, and essentially all of the GNN’s apparent permeability skill comes from the descriptor vector (which encodes size, lipophilicity, and hydrogen-bonding capacity—exactly the drivers of passive permeability). On the two toxicity endpoints the descriptors contribute a smaller but non-trivial margin (+0.08 AUC on hERG, +0.04 on Tox21); notably, the descriptor-free GNN on hERG (0.76) is comparable to the frozen ChemBERTa

baseline (0.770) rather than clearly superior to it. On the weakly-learnable clearance and half-life endpoints the effect is small or mixed, consistent with these endpoints being poorly predictable from structure-derived inputs of either kind. These results temper the “GNNs are best for structure-driven tasks” reading of Table IV: for the structure-driven endpoints the advantage is more accurately described as coming from the combined graph-plus-descriptor pipeline, and on Caco-2 predominantly from the descriptors.

D. Backbone Selection

A preliminary GNN architecture screen was conducted only to select a practical backbone for the main benchmark, not to establish a definitive ranking of GNN architectures. We evaluated eight common message-passing variants: GCN [8], GAT [43], GraphSAGE [44], GIN [9], TAG [45], SGC [46], TransformerConv [47], and GraphConv [40]. Each architecture was evaluated at its best of five configurations selected on the validation split (varying hidden dimension, number of layers, and learning rate) under the same scaffold-split pipeline used in the main study. Full per-architecture results for both the classification and regression endpoints (all eight backbones) are provided in the supplementary material (Tables S1 and S2, Figs. S1 and S2).

The screen shows that no single architecture dominates across endpoints. On the endpoints that are actually learnable from structure, several architectures cluster within a narrow

band: on Caco-2 the top architectures span $R^2 = 0.37$ – 0.49 (GraphConv 0.44), on Tox21 they span $AUC = 0.69$ – 0.76 (GraphConv 0.74, second best), and on hERG $AUC = 0.72$ – 0.80 (GraphConv 0.78). On the weakly-learnable clearance and half-life endpoints, by contrast, every architecture is unstable, with large negative R^2 values appearing for GCN, GIN, and GraphSAGE as well as GraphConv (Table S2 in the supplementary material); this instability is a property of the endpoints rather than of any single backbone, and is itself consistent with this paper’s central finding that predictability is endpoint-driven rather than architecture-driven. We therefore selected GraphConv as the backbone because it is consistently competitive on the learnable endpoints while being by far the most parameter-efficient candidate ($\approx 81k$ parameters versus 463k for GCN and 987k for TransformerConv) and among the fastest to train. This choice should not be interpreted as evidence that GraphConv is universally superior to other GNN architectures; architectures with a marginally better average regression rank (e.g., GCN and TransformerConv) require 5 – $12\times$ more parameters for differences that fall within the noise of these small, scaffold-split datasets. GraphConv thus served as a robust and computationally practical backbone for the controlled comparison of model families and hyperparameter optimization strategies reported in the main benchmark.

E. Effect of Optimization

Table VI presents the HPO algorithm comparison across all tasks, and Table VII summarizes the best configuration per dataset.

No single algorithm consistently dominates across all tasks. Random Search wins three of four regression tasks among NiaPy-based algorithms, while SA leads on Tox21 ($AUC = 0.742$) and ABC leads on hERG ($AUC = 0.825$). Table IX summarizes the paired performance differences between each algorithm and Random Search across the six datasets, using each task’s primary metric (RMSE for regression, F1 for classification; the per-optimizer classification F1 values are given in Table VIII) with a normalized direction so that positive values favor the optimizer. None of the metaheuristics achieves a net improvement over Random Search: the mean relative difference is negative for all five algorithms (between -3.9% and -7.0%), and the bootstrap 95% confidence interval either spans zero (PSO, GA) or lies entirely below it (ABC, SA, HC). Win/tie/loss counts tell the same story, with every algorithm losing on a majority of datasets and HC losing on all six. This indicates that, under our 50-trial budget, optimization strategy has limited impact compared to task characteristics, and that Random Search is a strong baseline rather than one that adaptive search reliably surpasses.

The training-convergence visualization (Fig. 5) illustrates the qualitative per-epoch convergence of the best configuration selected by each algorithm: PSO and ABC reach strong validation values early, while SA improves more gradually during training.

The competitiveness of Random Search on regression tasks is consistent with the “effective dimensionality” phenomenon [25]: when only a subset of hyperparameters (e.g.,

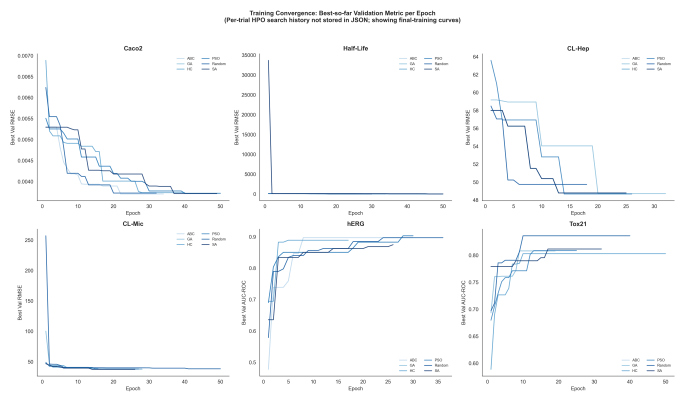


Fig. 5. Convergence trajectories for HPO algorithms across all six benchmark datasets (four ADME regression and two toxicity classification tasks), showing the best-so-far validation metric per training epoch for the best configuration selected by each algorithm. The curves illustrate qualitative optimization behavior recorded during the HPO analysis.

learning rate, hidden dimension) dominates performance, the probability of Random Search sampling a near-optimal configuration by chance increases. To test whether this holds in our own search space rather than by analogy, we ran a dedicated random search of 80 configurations over the full seven-dimensional space (Table II) on a representative learnable endpoint (Caco-2), logging the validation R^2 of every configuration, and attributed performance variance to each hyperparameter using permutation importance from a random-forest surrogate (out-of-bag $R^2 = 0.72$) together with Spearman rank correlations. Two of the seven dimensions—learning rate and hidden dimension—account for 81% of the surrogate importance (learning rate 42%, Spearman $\rho = 0.56$; hidden dimension 40%, $\rho = -0.53$; both $p < 0.001$), with weight decay contributing a further 15% and the three MLP-head widths and the number of layers together under 5%. This concentration of variance in a low-dimensional effective subspace is exactly the condition under which Random Search is expected to be competitive, and it converts the Bergstra–Bengio explanation from an analogy into a finding demonstrated on this study’s own trials. On the small hERG endpoint the same sweep yields a much noisier surrogate (out-of-bag $R^2 < 0$, validation AUC spanning only 0.74–0.94), with learning rate again the single most influential dimension; the weak surrogate is itself consistent with the seed-level instability documented for this dataset.

Fig. 9 compares all algorithms across datasets on test RMSE, R^2 , and training time; the performance-time panel shows that Random Search and PSO reach competitive accuracy at low computational cost.

All HPO algorithms achieve $AUC-ROC > 0.65$ on both classification tasks. On hERG, the best model (ABC) achieves $F1 = 0.809$ with balanced performance across both classes. On Tox21, the best model (SA) achieves $F1 = 0.455$ despite 96.2% accuracy, reflecting the severe class imbalance (4.9% positive rate on the scaffold-split test set; $\approx 4.2\%$ across the full corpus). Per-class confusion matrices for both best models are shown in Fig. 6.

TABLE VI

HPO ALGORITHM COMPARISON ACROSS DATASETS. VALUES SHOW TEST METRICS FOR EACH OPTIMIZER. BEST PER DATASET IN **BOLD**. TPE OPERATES OVER A SLIGHTLY EXPANDED SEARCH SPACE THAT INCLUDES DROPOUT.

Dataset	PSO	ABC	GA	SA	HC	Random	TPE
<i>Regression (RMSE ↓)</i>							
Caco2_Wang	0.0031	0.0029	0.0031	0.0029	0.0030	0.0027	0.0029
Half_Life_Obach	21.66	21.66	21.66	23.70	24.52	22.31	21.48
Clearance_Hep.	70.21	72.04	71.34	72.04	72.04	68.22	80.32
Clearance_Mic.	42.76	42.29	42.29	40.94	41.63	38.75	40.89
<i>Classification (AUC-ROC ↑)</i>							
Tox21 (NR-AR)	0.692	0.735	0.735	0.742	0.652	0.713	0.722
hERG	0.747	0.825	0.747	0.802	0.821	0.747	0.756

TABLE VII

BEST TEST SET PERFORMANCE PER DATASET AMONG NIAPY-BASED HPO ALGORITHMS (SINGLE BEST TRIAL). FOR THE REGRESSION ROWS THE METRIC COLUMNS REPORT RMSE AND R^2 ; FOR THE CLASSIFICATION ROWS THEY REPORT F1 AND AUC-ROC. “TIME (s)” IS THE WALL-CLOCK TIME TO TRAIN THE SINGLE BEST CONFIGURATION TO CONVERGENCE, NOT A REPRESENTATIVE PER-TRIAL COST; THE TOX21 VALUE (1039.6 s) REFLECTS THE LARGEST DATASET TRAINED UNDER THE FULL EPOCH BUDGET. WHEN SEVERAL ALGORITHMS CONVERGE TO THE SAME BEST CONFIGURATION AND HENCE IDENTICAL METRICS (E.G., PSO, ABC, AND GA ON HALF-LIFE ALL REACH RMSE = 21.66), THE EARLIEST IN THE OPTIMIZER ENUMERATION ORDER (PSO, ABC, GA, SA, HC) IS REPORTED.

Dataset	Best Algo	RMSE/F1	R^2 /AUC	Time (s)
Caco2_Wang	Random	0.0027	0.481	15.7
Half_Life	PSO	21.66	0.004	19.7
Clear_Hep.	Random	68.22	−1.019	5.3
Clear_Mic.	Random	38.75	0.191	52.3
Tox21 (NR-AR)	SA	0.455	0.742	1039.6
hERG	ABC	0.809	0.825	33.9

TABLE VIII

PER-OPTIMIZER TEST F1 ON THE TWO CLASSIFICATION TASKS (SINGLE BEST TRIAL), REPORTED ALONGSIDE THE AUC-ROC OF TABLE VI.

THESE ARE THE F1 VALUES ENTERING THE PAIRED COMPARISON AGAINST RANDOM SEARCH (TABLE IX), WITH RANDOM SEARCH AS THE BASELINE. TPE IS OMITTED BECAUSE IT OPTIMIZES A DIFFERENT SEARCH SPACE AND IS EXCLUDED FROM THE PAIRED COMPARISON.

Dataset	PSO	ABC	GA	SA	HC	Random
Tox21 (NR-AR)	0.430	0.463	0.463	0.455	0.433	0.469
hERG	0.857	0.809	0.857	0.859	0.830	0.833

TABLE IX

HPO ALGORITHMS VS. RANDOM SEARCH: WIN/TIE/LOSS, MEAN RELATIVE IMPROVEMENT AND BOOTSTRAP 95% CONFIDENCE INTERVALS ACROSS $n=6$ DATASETS. IMPROVEMENT DIRECTION IS NORMALIZED SO THAT POSITIVE FAVORS THE OPTIMIZER (RMSE: RANDOM—ALGO; F1: ALGO—RANDOM), EXPRESSED AS A PERCENTAGE OF THE

RANDOM-SEARCH BASELINE. CIs ARE FROM 10,000 PERCENTILE BOOTSTRAP RESAMPLES OF THE SIX PAIRED DIFFERENCES; d_z IS THE PAIRED EFFECT SIZE (MEAN/STD OF THE DIFFERENCES).

Algorithm	W/T/L	Mean Δ (%)	95% CI (%)	d_z
PSO	2/0/4	−5.11	[−10.27, +0.04]	−0.70
ABC	1/0/5	−3.92	[−6.96, −0.49]	−0.89
GA	2/0/4	−4.01	[−9.22, +0.92]	−0.57
SA	1/0/5	−4.14	[−6.41, −1.06]	−1.09
HC	0/0/6	−7.04	[−9.54, −4.03]	−1.86

Confusion Matrices for Best GNN Classification Models

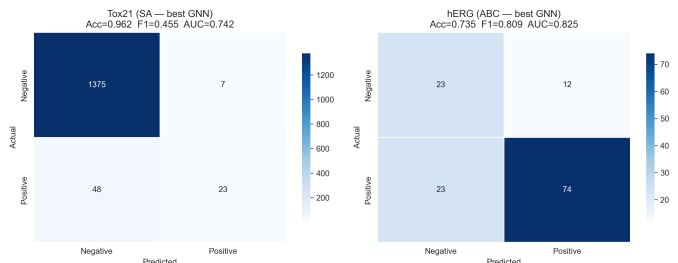


Fig. 6. Confusion matrices for the best-performing toxicity classification models. (A) Tox21 shows majority-class bias due to extreme imbalance (SA, F1 = 0.455). (B) hERG shows balanced performance (ABC, F1 = 0.809).

F. Imbalance-Aware Metrics and Calibration

Because ROC-AUC and accuracy can be misleading under strong class imbalance, we report two additional metrics that are sensitive to minority-class behavior: the area under the precision-recall curve (PR-AUC, equivalently average precision) and the Matthews correlation coefficient (MCC) [48]. PR-AUC is threshold-free and should be compared against the positive-class prevalence, which is the score of a random classifier; the prevalence is 0.049 for Tox21 (NR-AR) and 0.735 for hERG on the scaffold-split test set. MCC and F1 are threshold-dependent, so for these two metrics we select the decision threshold that maximizes MCC on the validation split and apply it unchanged to the test split, a standard operating-point selection that never uses test labels. All values are reported as mean $\pm$ standard deviation over the five seeds [42, 123, 456, 789, 1011] using the best HPO-selected configuration per task (Table VII); they complement the ROC-AUC reported in Table XI. We deliberately report these bounded, minority-class metrics as mean $\pm$ standard deviation, rather than the median [min–max] used for the ROC-AUC in Table XI, because here the across-seed spread is itself the quantity of interest: the standard deviation directly conveys how unstable minority-class performance becomes under class imbalance. The ROC-AUC of Table XI, by contrast, is summarized by the median precisely to prevent a single divergent seed from dominating a central-tendency estimate.

Table X reports these imbalance-aware metrics. For both endpoints the PR-AUC exceeds the prevalence baseline, indicating that the models recover genuine minority-class signal rather than exploiting the majority class: Tox21 reaches PR-

TABLE X

IMBALANCE-AWARE TOXICITY METRICS (MEAN $\pm$ STD OVER $n=5$ SEEDS). PR-AUC IS THRESHOLD-FREE; F1 AND MCC USE A DECISION THRESHOLD SELECTED ON THE VALIDATION SPLIT TO MAXIMIZE MCC. FOR REFERENCE, THE PR-AUC OF A RANDOM CLASSIFIER EQUALS THE POSITIVE PREVALENCE (0.049 FOR Tox21, 0.735 FOR hERG).

Dataset	PR-AUC	F1	MCC
Tox21 (NR-AR)	0.168 $\pm$ 0.108	0.243 $\pm$ 0.118	0.219 $\pm$ 0.142
hERG	0.876 $\pm$ 0.035	0.865 $\pm$ 0.034	0.470 $\pm$ 0.077

AUC = 0.168 against a 0.049 baseline (a $3.4\times$ improvement), and hERG reaches 0.876 against a 0.735 baseline. The MCC values tell a more nuanced story. hERG attains a moderate MCC of 0.470, whereas Tox21 attains only 0.219 with a large across-seed spread, consistent with the difficulty of the $\approx 4.9\%$ positive rate. A notable failure mode is that at the default 0.5 threshold three of the five Tox21 seeds collapse entirely to the majority class (F1 = MCC = 0); only after validation-based threshold selection do they yield non-degenerate minority-class predictions. This confirms that for severely imbalanced safety endpoints the choice of operating point is as consequential as the choice of model, and that a single threshold-free imbalance-aware metric such as PR-AUC is essential for fair comparison.

Fig. 7 summarizes the across-seed variability of these metrics. The error bars are tight for hERG but wide for Tox21 (for example, PR-AUC 0.168 ± 0.108), showing that prediction quality on the heavily imbalanced endpoint is not only lower on average but also substantially less stable across random initializations. These threshold- and rank-based metrics remain stable for hERG even though the AUC-based multi-seed validation (Table XI) shows one divergent seed, because on the majority-positive hERG endpoint F1 and PR-AUC are far less sensitive to a single low-AUC run than ROC-AUC is. Fig. 8 shows the corresponding reliability diagrams, where the predicted probabilities are pooled over the five seeds and grouped into ten equal-width bins. Both models are reasonably well calibrated, with a low Expected Calibration Error (ECE = 0.020 for Tox21 and 0.067 for hERG); the hERG curve lies slightly below the diagonal at mid-range confidences, indicating mild over-confidence that is relevant when predicted probabilities are used directly for risk ranking.

G. Effect of Evaluation Protocol

a) Scaffold split compresses performance differences.:

Table XI presents multi-seed validation results for the best HPO-selected configurations, reported as the median and full range across the five seeds. Across-seed stability varies by endpoint and does not simply track structure-drivenness: Caco-2, Hepatocyte clearance, and Tox21 vary within a narrow band, whereas the best configuration is markedly less stable on Half-Life and Microsomal clearance, where individual seeds diverge to much larger RMSE, and on hERG one of the five seeds collapses toward chance (AUC = 0.64) while the others remain near 0.80. We report the median rather than the mean because the median is robust to these occasional divergent runs; the wide ranges reinforce our central finding

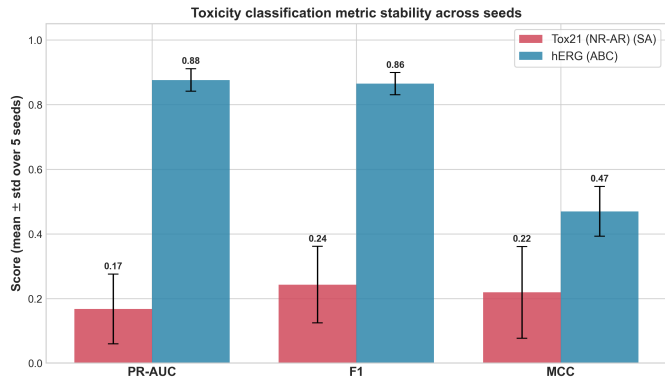


Fig. 7. Across-seed stability of the imbalance-aware toxicity metrics (PR-AUC, F1, MCC). Bars show the mean over five seeds and error bars show $\pm$ one standard deviation. Variability is small for hERG but large for Tox21, reflecting the instability induced by extreme class imbalance.

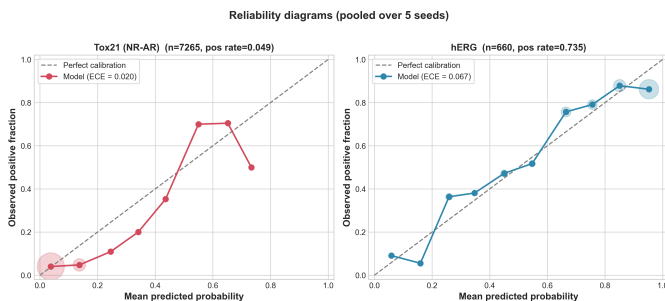


Fig. 8. Reliability diagrams for the toxicity classifiers, with predicted probabilities pooled over five seeds and binned into ten equal-width intervals. Bubble size is proportional to the number of test molecules in each bin. Both models track the diagonal closely (ECE = 0.020 for Tox21, 0.067 for hERG), indicating reasonably calibrated probabilities.

that the weakly-learnable pharmacokinetic endpoints do not support robust structure-only prediction. The single divergent hERG seed is most consistent with an initialization-dependent training instability on this small 590-compound scaffold-split dataset rather than a systematic failure of the endpoint: the other four seeds remain tightly clustered near AUC = 0.80, and the threshold- and rank-based metrics of Table X stay stable across all five seeds. This behavior is a practical caution that even a structure-driven, GNN-suitable endpoint can require multi-seed retraining to obtain a dependable performance estimate on small datasets. The multi-seed medians for the two toxicity endpoints (hERG AUC median = 0.796, Tox21 AUC median = 0.710) fall slightly below the single best-trial figures reported in Table IV (0.825 and 0.742), because selecting the best of 50 HPO trials is itself an optimistic estimator; the headline single-trial values should therefore be read as best-case point estimates and the multi-seed medians as the more conservative summary. On Caco-2 the two estimates agree closely (multi-seed median RMSE 0.0027 versus best-trial 0.0027).

b) Evaluation protocol influences model rankings.:

Scaffold-based splitting creates a structural covariate shift where the model must generalize not merely to unseen molecules, but to unseen chemical families. Under such conditions, validation performance becomes a noisier proxy for

TABLE XI

MULTI-SEED VALIDATION RESULTS USING THE BEST HPO CONFIGURATION PER DATASET, REPORTED AS THE MEDIAN [MIN-MAX] ACROSS FIVE SEEDS [42, 123, 456, 789, 1011]. THE MEDIAN IS USED BECAUSE THE BEST CONFIGURATION DIVERGES ON A MINORITY OF SEEDS FOR THE WEAKLY-LEARNABLE ENDPOINTS (HALF-LIFE, MICROSOMAL CLEARANCE) AND FOR ONE hERG SEED, MAKING THE MEAN UNREPRESENTATIVE; THE WIDE RANGES QUANTIFY THIS ACROSS-SEED INSTABILITY.

Dataset	Task	Metric	Median [min-max], $n=5$
Caco2	Regr.	RMSE	0.0027 [0.0025–0.0030]
Half_Life	Regr.	RMSE	22.74 [20.74–49.38]
Clear_Hep.	Regr.	RMSE	49.37 [48.17–52.46]
Clear_Mic.	Regr.	RMSE	50.38 [39.58–106.67]
Tox21	Class.	AUC	0.710 [0.703–0.725]
hERG	Class.	AUC	0.796 [0.644–0.815]

true generalization, because a configuration that happens to fit the specific scaffolds in the validation set may not transfer to the test scaffolds. This has important consequences for model selection: on Tox21, ChemBERTa achieves validation AUC = 0.896, but drops to 0.73 on the scaffold-split test set, while MolCLR falls below the random baseline on the test set with AUC = 0.45. These results demonstrate that evaluation protocol substantially influences model rankings and perceived performance.

c) Fine-tuning does not rescue the pretrained encoder.:

Because our main comparison uses ChemBERTa as a frozen feature extractor, we tested whether the more favorable fine-tuning setting closes the gap to the GNN on the two toxicity endpoints. We fine-tuned ChemBERTa end-to-end (unfreezing the last two transformer layers, with a class-weighted loss for the imbalanced Tox21 task) under the same scaffold split. Fine-tuning did not improve either endpoint: on hERG the test AUC was 0.771, essentially identical to the frozen baseline (0.770) and still below the GNN (0.825); on Tox21 it collapsed to 0.474, below the frozen baseline (0.728) and below random, consistent with a transformer overfitting the small, heavily imbalanced scaffold-split training set. The GNN’s advantage over ChemBERTa on these endpoints is therefore not an artifact of the frozen evaluation setting; if anything, fine-tuning a large pretrained encoder is counterproductive under scaffold shift on these small toxicity datasets.

The noise introduced by scaffold splitting also explains why within the evaluated datasets and 50-trial budget, we did not observe consistent evidence that the evaluated HPO strategies outperform Random Search (Table IX), with paired confidence intervals that span or fall below zero. Adaptive methods use validation performance to guide search decisions, when that signal is noisy, they risk optimizing for the specific validation fold rather than discovering genuinely superior hyperparameters. Random Search, by contrast, does not condition on previous evaluations, making it less susceptible to sequential overfitting of a noisy validation surface.

V. DISCUSSION

Predictive performance varied widely across the six endpoints, from $R^2 = 0.48$ on Caco-2 permeability to $R^2 = -1.02$

on hepatocyte clearance, confirming that ADMET prediction is strongly endpoint-dependent. Across the evaluated datasets, model rankings were not consistent: architectures that performed well on permeability or toxicity did not necessarily transfer to clearance or half-life prediction. This indicates that ADMET modeling should not be treated as a single homogeneous prediction problem. Instead, each endpoint reflects a different combination of physicochemical, biological, and experimental factors, which affects how much predictive signal can be extracted from molecular structure alone.

For endpoints such as Caco-2 permeability and hERG cardiotoxicity, graph-based models achieved competitive performance, suggesting that molecular topology and local chemical environments contain useful predictive information. This is expected because permeability and ion-channel inhibition are influenced by structural properties such as molecular size, lipophilicity, charge distribution, hydrogen-bonding capacity, and specific functional groups. In these cases, GNNs can benefit from atom-level and bond-level message passing, allowing them to capture substructural patterns that are directly relevant to the endpoint.

In contrast, Half-Life and Hepatocyte clearance showed weak predictive performance across model families, with near-zero or negative R^2 values. This suggests that these tasks depend on biological and experimental factors not encoded in molecular structure alone, including enzyme-mediated metabolism, transporter interactions, tissue distribution, and plasma protein binding effects [49]–[51]. The poor performance on these endpoints is therefore better explained by limitations of structure-only molecular representations than by architecture alone. In our experiments, increasing model complexity did not compensate for the apparent lack of biological context in the structure-only input representation, consistent with the information required to explain these targets not being captured by molecular structure alone.

The classification results further demonstrate the importance of endpoint-specific evaluation. Although toxicity and cardiotoxicity tasks are both classification problems, they differ substantially in class balance and biological mechanism. For example, the Tox21 NR-AR task is highly imbalanced, which can bias standard training objectives toward the majority class [33]. This makes ROC-AUC more appropriate than accuracy for comparing classifiers, but it also means that high AUC values should be interpreted carefully when the positive class is rare. Future work should therefore evaluate additional metrics such as precision-recall AUC, sensitivity, specificity, and calibration error, particularly for safety-related endpoints where false negatives can be costly.

The scaffold-split protocol also plays an important role in interpreting the results. Compared with random splitting, scaffold splitting provides a stricter evaluation setting because compounds with the same core scaffold are assigned to the same partition. This reduces the risk that models exploit close structural analogues in the training set and gives a more realistic estimate of performance on novel chemical series. The observed performance drops under scaffold-split evaluation therefore highlight the difficulty of generalizing beyond familiar chemotypes, which is a central challenge in

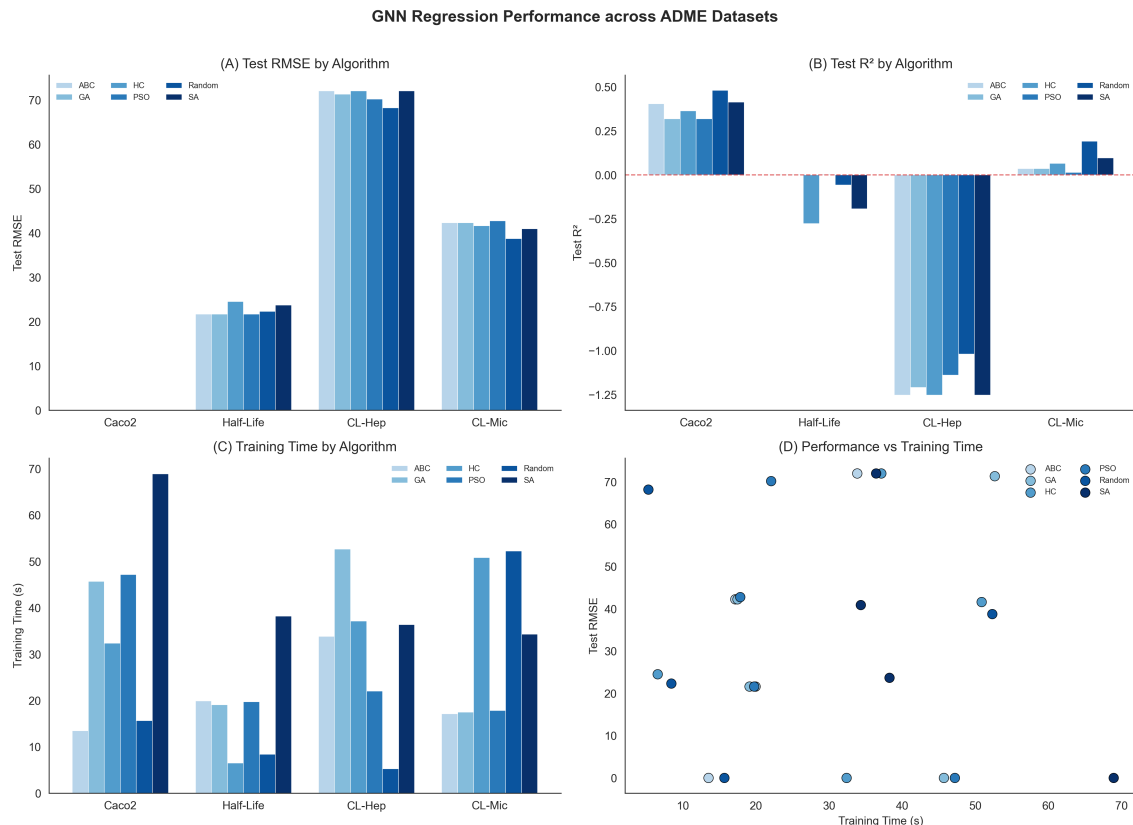


Fig. 9. HPO algorithm performance comparison across ADME datasets. (A) Test RMSE by algorithm, (B) Test R^2 scores, (C) Training time comparison, (D) Performance vs. training time trade-off.

practical drug discovery.

The comparison between GNNs, fingerprints, descriptors, and pretrained molecular representation models suggests that no single representation is uniformly optimal. The frozen pretrained encoders evaluated here can provide useful molecular embeddings, but their performance depends on the alignment between the pretraining objective, the molecular representation, and the downstream endpoint. Similarly, classical fingerprint-based models remain competitive for some endpoints, especially when the dataset is small or when the target is strongly associated with known physicochemical properties. These results support endpoint-specific model selection rather than the use of a single architecture across all ADMET tasks.

The hyperparameter optimization results also suggest that optimization should be interpreted cautiously. None of the evaluated metaheuristics improved on Random Search on average: their paired 95% confidence intervals against Random Search span or fall below zero, and the mean relative difference is negative for every algorithm. Where individual algorithms win on a particular dataset, the advantage is small and does not transfer across datasets or architectures. For small and noisy ADMET datasets, validation performance may be unstable, and aggressive optimization can overfit to the validation split. This reinforces the need for repeated runs, consistent evaluation protocols, and transparent reporting of variance across seeds. In practice, simple optimization strategies may be sufficient for some endpoints, while more

expensive search methods should be justified by consistent improvements across independent runs.

These results support endpoint-specific model selection. Structure-based models are most appropriate for endpoints with clear structure-activity relationships, whereas clearance and half-life prediction likely require additional biological or assay-level variables, including enzyme activity, transporter information, protein binding, and experimental conditions.

In biomedical informatics workflows, ADMET predictors should therefore be interpreted as endpoint-specific decision-support tools rather than interchangeable screening modules. Structure-only models may be useful for hERG and Caco-2 screening, but their predictions for hepatocyte clearance and half-life should be treated cautiously unless additional biological context is available.

This study also has several limitations that bound the scope of our conclusions. First, the benchmark focuses on six TDC ADMET endpoints, which cover important but not exhaustive aspects of drug safety and pharmacokinetics. Second, the pretrained encoders (ChemBERTa, MolCLR) are evaluated primarily in a frozen feature-extraction setting rather than under full task-specific fine-tuning; for the two toxicity endpoints we additionally fine-tuned ChemBERTa and found that it did not improve over the frozen baseline (hERG) and collapsed below random on the small, imbalanced Tox21 set (Section IV-G), so the frozen setting does not appear to understate its performance on these datasets, though a broader fine-tuning

study across endpoints and encoders remains future work. The pretrained encoders are also treated as representative rather than exhaustive examples of pretrained molecular models. Third, all model families operate on structure-only inputs, so biological and experimental covariates (assay conditions, metabolizing enzymes, transporter and protein-binding measurements) are not included. We also note that the end-to-end GNN receives, in addition to its learned graph representation, a small vector of structure-derived physicochemical descriptors that the frozen pretrained encoders and fingerprint baselines do not; although these descriptors carry no biological or assay information, the model-family comparison is therefore not strictly input-matched. The controlled descriptor ablation in Section IV-C (Table V) quantifies this asymmetry: it accounts for essentially all of the GNN’s Caco-2 performance and a smaller +0.08 AUC on hERG, so part of the GNN’s advantage on the structure-driven endpoints reflects this auxiliary input rather than the graph representation alone. Fourth, scaffold splitting is a more realistic evaluation protocol than random splitting, but it is still a retrospective benchmark and not a full prospective validation on newly synthesized chemical series. Fifth, the datasets are small (hundreds to a few thousand compounds) and, for the toxicity tasks, class-imbalanced (e.g., a $\approx 4.2\%$ positive rate on Tox21 NR-AR), which increases variance across splits and can bias training objectives. Finally, the hyperparameter optimization comparison is limited to a 50-trial budget per algorithm and dataset; a larger budget could alter the relative standing of the metaheuristics against Random Search. Accordingly, our conclusions apply within the evaluated datasets, model families, scaffold-split protocol, and frozen/pretrained setup, and further validation on additional datasets and prospective experimental settings would strengthen them.

The benchmark can be extended in three directions. First, additional ADMET endpoints from the TDC benchmark and external pharmaceutical datasets could be included to test whether the observed endpoint-dependent trends generalize more broadly. Second, multimodal models that combine molecular graphs with assay metadata, protein targets, enzyme information, or transcriptomic features could be evaluated for endpoints where structure-only representations fail. Third, uncertainty estimation and calibration should be incorporated to assess not only predictive accuracy but also the reliability of model confidence, which is especially important for safety-critical ADMET applications.

VI. CONCLUSION

In this work, we investigated the task-dependent performance of molecular graph neural networks and pretrained representations for ADMET prediction under scaffold-split evaluation. Through approximately 2,370 training runs and controlled comparisons across five model families and six TDC ADMET benchmarks, our results show that the usefulness of structure-only molecular models depends strongly on the endpoint being predicted. Structure-only models achieved strong performance on endpoints with clearer structure-activity relationships, such as hERG cardiotoxicity (best-trial AUC =

0.825; multi-seed median 0.796) and Caco-2 permeability ($R^2 = 0.48$). In contrast, the same modeling approaches failed to provide reliable predictions for pharmacokinetic endpoints such as Half-Life and Hepatocyte Clearance, where both GNNs and pretrained encoders yielded near-zero or negative R^2 within the evaluated benchmark.

These results support the main conclusion of this study, that ADMET model selection should be endpoint-specific. Molecular structure is informative for some endpoints, but insufficient for others. We identify three levels of task learnability from molecular structure. First, structure-driven tasks, such as hERG and Caco-2, where GNNs match or outperform the frozen pretrained encoders evaluated here and achieve practically useful predictions. Second, moderately structure-driven tasks, such as Tox21 NR-AR and Microsomal Clearance, where GNNs still achieve the best performance but the margin over the other model families narrows considerably. Third, weakly structure-driven tasks, such as Half-Life and Hepatocyte Clearance, where all evaluated models perform poorly regardless of architecture, pretraining, or optimization strategy. This demonstrates that increasing model complexity is beneficial only when the endpoint contains sufficient structure-derived signals.

Scaffold-split evaluation is critical for drawing realistic conclusions from ADMET benchmarks. Random splits can produce overly optimistic estimates by allowing structurally similar molecules to appear in both training and test sets. In contrast, scaffold splitting provides a stricter assessment of generalization to unseen chemical scaffolds. Under scaffold-split conditions, performance differences between model families and hyperparameter optimization strategies become smaller, and model rankings can change. We further show that no hyperparameter optimization algorithm consistently outperforms Random Search at a 50-trial budget, with paired 95% confidence intervals that span or fall below zero. This reinforces that evaluation protocol strongly influences the apparent benefit of both model architecture and optimization methods.

This study has several limitations. First, all evaluated models operate on structure-only molecular representations and do not incorporate biological, experimental, or pharmacokinetic context. This likely contributes to the weak performance observed on Half-Life and Hepatocyte Clearance. Second, pretrained models were evaluated primarily in a frozen setting; a fine-tuning check on the two toxicity endpoints did not improve on the frozen ChemBERTa baseline (and degraded on the imbalanced Tox21 set), but a broader fine-tuning study across all endpoints and encoders remains future work. Third, the conclusions are based on six benchmark datasets of limited size, and results may vary under larger or more diverse ADMET datasets.

Future work should move beyond structure-only prediction by incorporating additional sources of information, including protein targets, assay conditions, experimental metadata, dose-dependent effects, and pharmacokinetic context. Such information may be especially important for endpoints where molecular structure alone is weakly predictive. Further work should also investigate uncertainty-aware model selection and evaluation protocols that better reflect prospective drug dis-

covery settings.

Overall, the strongest contribution of this study is showing that ADMET prediction should not be treated as a single uniform modeling problem. Structure-only models can be effective for endpoints such as hERG and Caco-2, but they fail on endpoints such as Half-Life and Hepatocyte Clearance under scaffold-split evaluation. Therefore, reliable ADMET modeling requires endpoint-specific model selection and realistic scaffold-based evaluation. This supports a cautious use of molecular AI models as decision-support tools in early-stage drug discovery rather than as universally reliable predictors across all ADMET endpoints.

DATA AND CODE AVAILABILITY

All datasets used in this study are publicly available through the Therapeutics Data Commons (TDC). The source code, trained model configurations, hyperparameter-optimization logs, and experimental results are available at:

<https://github.com/NitramVonemats/ADMET-Benchmark>

Multi-seed validation used the fixed seeds [42, 123, 456, 789, 1011], and each HPO algorithm was run for 50 trials per dataset. Software dependencies and exact versions are documented in the repository, including Python ≥ 3.8 , PyTorch ≥ 2.0 , PyTorch Geometric, RDKit, PyTDC, NiaPy, Optuna, and Transformers.

ETHICAL COMPLIANCE

This study uses only publicly available benchmark datasets from the Therapeutics Data Commons and does not involve human subjects, identifiable patient data, animal experiments, or newly collected clinical data. Therefore, institutional review board approval and informed consent were not required.

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